

Total messages:

1 867 000

→

867k

+

120k

+

24.1k

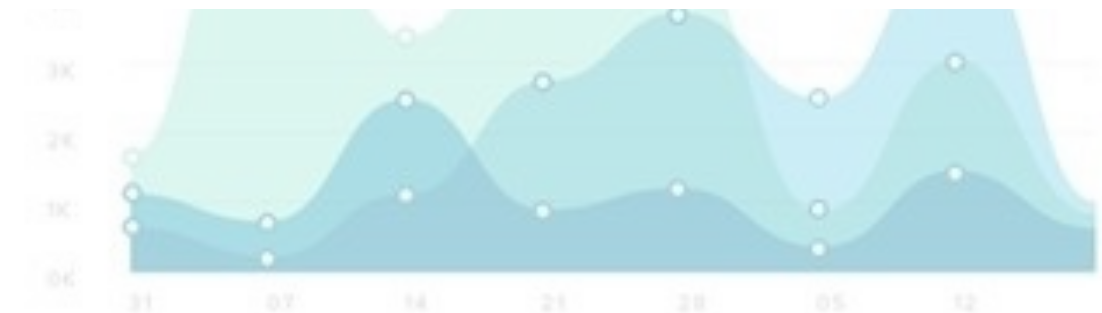
SSL	NON-SSL
↑ 652k	↑ 20k

SSL	NON-SSL
↑ 440k	↑ 0

SSL	NON-SSL
↑ 40k	↑ 8k

SSL	NON-SSL
↓ 7.1k	↑ 12k

DATA WAREHOUSE CONCEPTS, DESIGN & ARCHITECTURE



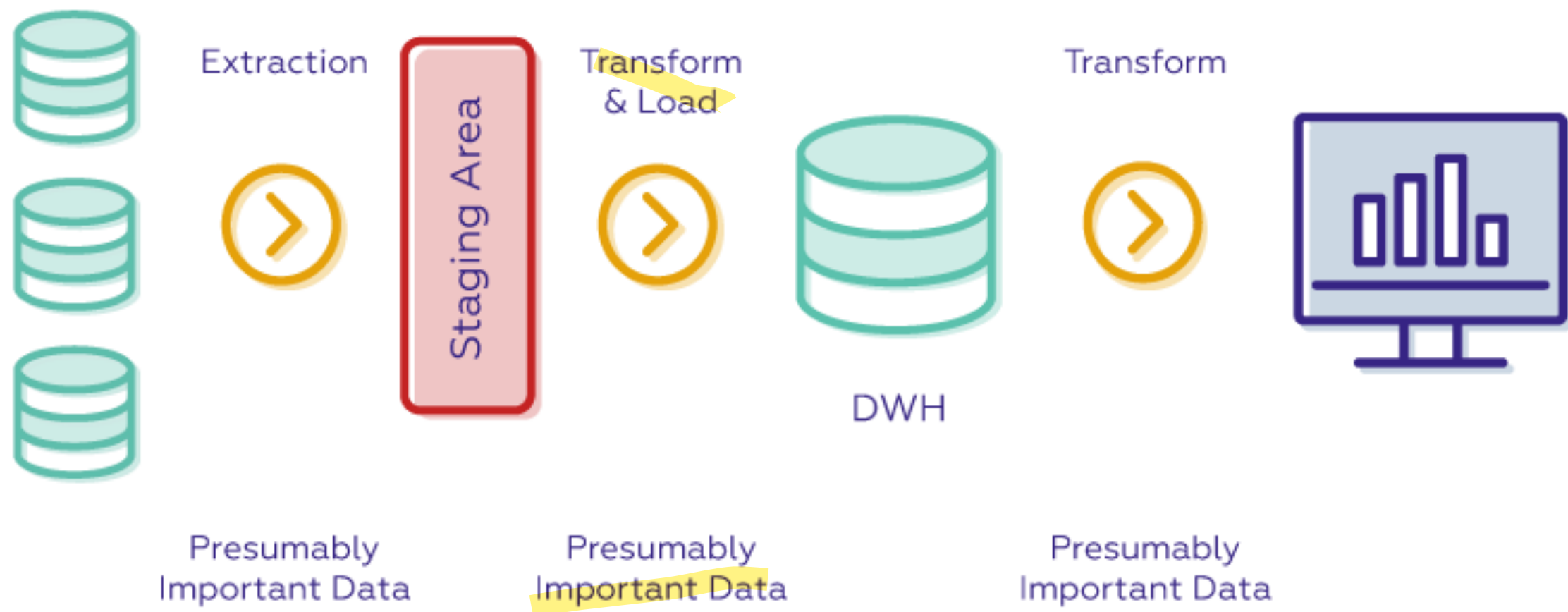
DATA WAREHOUSE

Architectural Design: ETL

- To serve its purpose of facilitating business analysis, the data warehouse must be loaded regularly
- **Data integration** is a challenge in data warehouse environments due to the necessity of integrating, rearranging and consolidating large volumes of data over many systems
- The process of extracting data from source systems and bringing it into the data warehouse is commonly called **ETL** (Extract, Transform, and Load)
- The main actions in the process in the ETL process are: Extraction of Data and Transportation of Data

DATA WAREHOUSE

Architectural Design: ETL



DATA WAREHOUSE

Architectural Design: ETL

- Extraction is the operation of extracting data from a **Source systems**
- Source systems range from typically transaction processing applications, to data sourced or exchanged externally with customers, prospects, suppliers, partners, and other stakeholders
- Designing and creating the extraction process is often one of the most time-consuming tasks in the ETL process (complexity of source systems, lack of documentation, data quality issues...)
- Data Extraction **is a continuous process that has to be executed periodically** to keep the data warehouse up-to-date

DATA WAREHOUSE

Architectural Design: ETL

- The Data Extraction method highly depends on the source system and the business needs in the target data warehouse environment
- Data extraction can be **full** or **incremental**.
- In full extraction, data is extracted completely from the source system
- In incremental extraction, only the data that has changed since a well-defined event back in history is extracted. It requires a mechanism to identify the delta change
- In many data warehouses, entire tables from the source systems are extracted to the data warehouse or staging area, and compared with a previous extract from the source system to identify the changed data

DATA WAREHOUSE

Architectural Design: ETL

- **Change tracking** is the key-enabling technology for providing near real-time, or on-time, data warehousing
- Timestamps can be added to tables in the staging area to keep track of last loaded records
- Data can be extracted from the source system **online** or **offline**
- **Online extraction** means data is extracted **directly from the source** system itself into the data warehouse or an intermediate system
- In **Offline extraction**, data is staged explicitly outside the original source system (e.g. flat files, archive logs, db dump...)

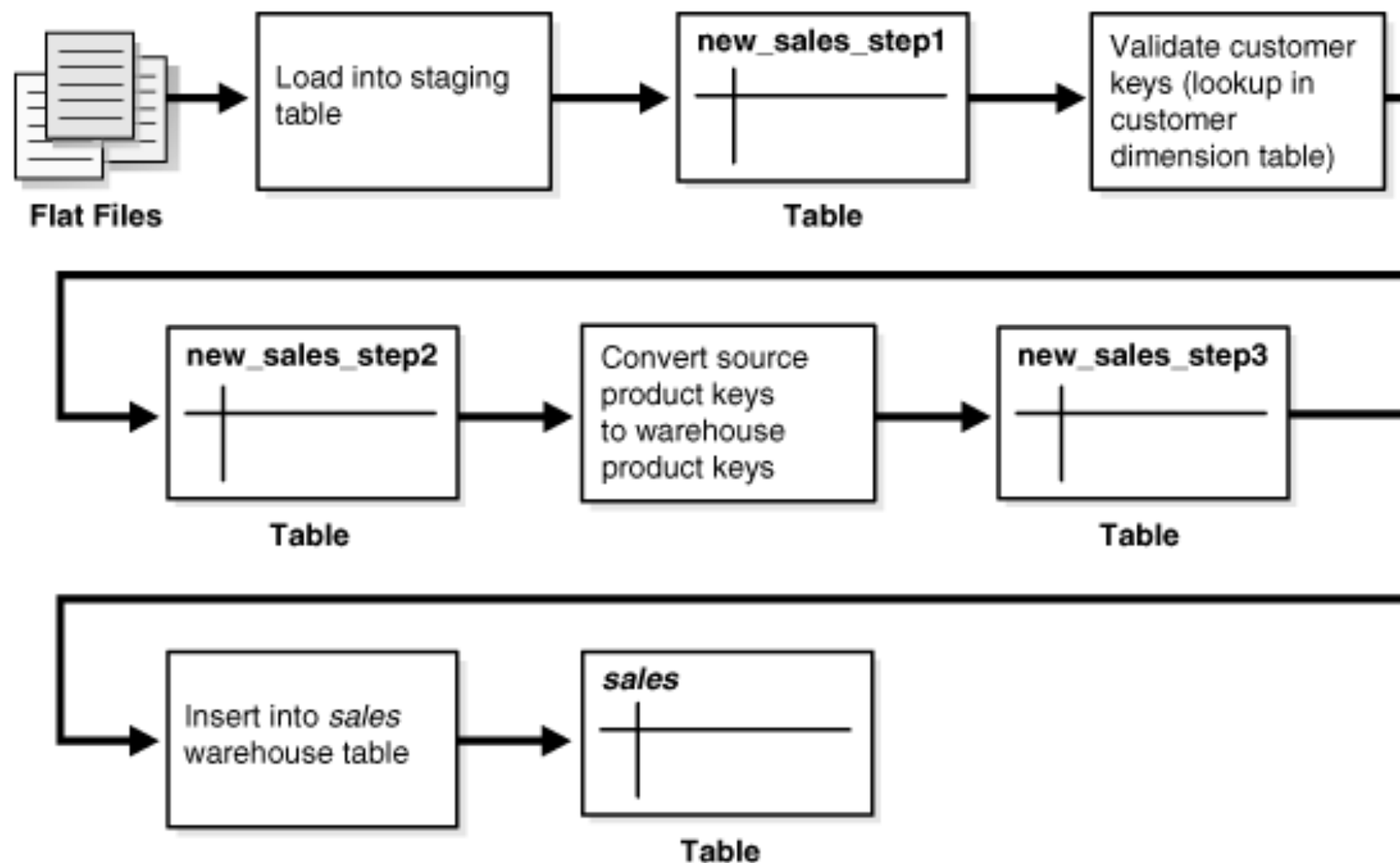
DATA WAREHOUSE

Architectural Design: ETL

- Data transformations are often the most complex and, in terms of processing time, the most costly part of the ETL process
- Transformations can range from simple data conversions to extremely complex data scrubbing techniques
- Transformations are generally pipelined into a series of processing steps that improve the quality of the data received from the source, and merge data from two or more sources to create and enforce conformed dimensions and conformed metrics
- ETL tools, rather than custom code, are used to help manage the complex process

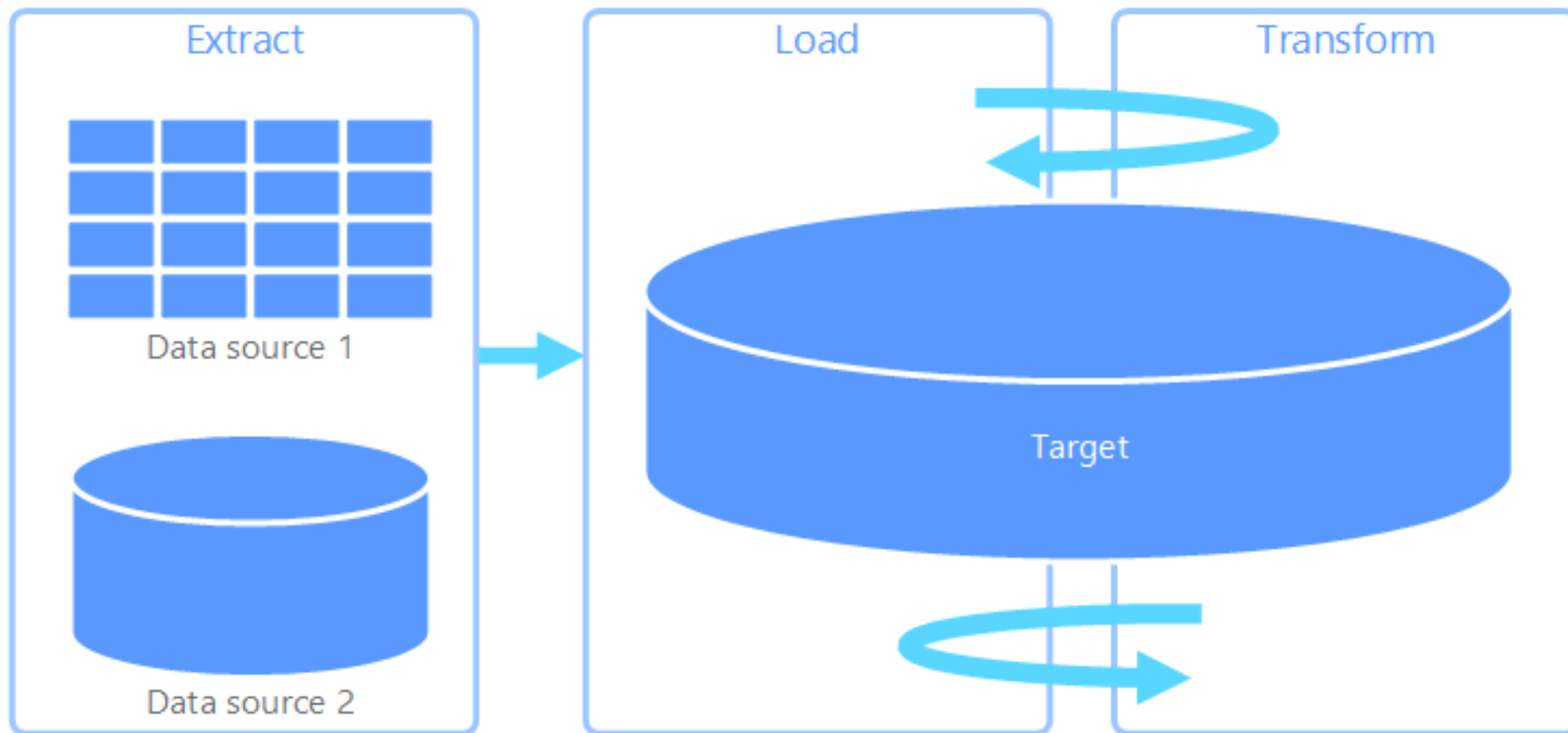
DATA WAREHOUSE

Architectural Design: ETL



DATA WAREHOUSE

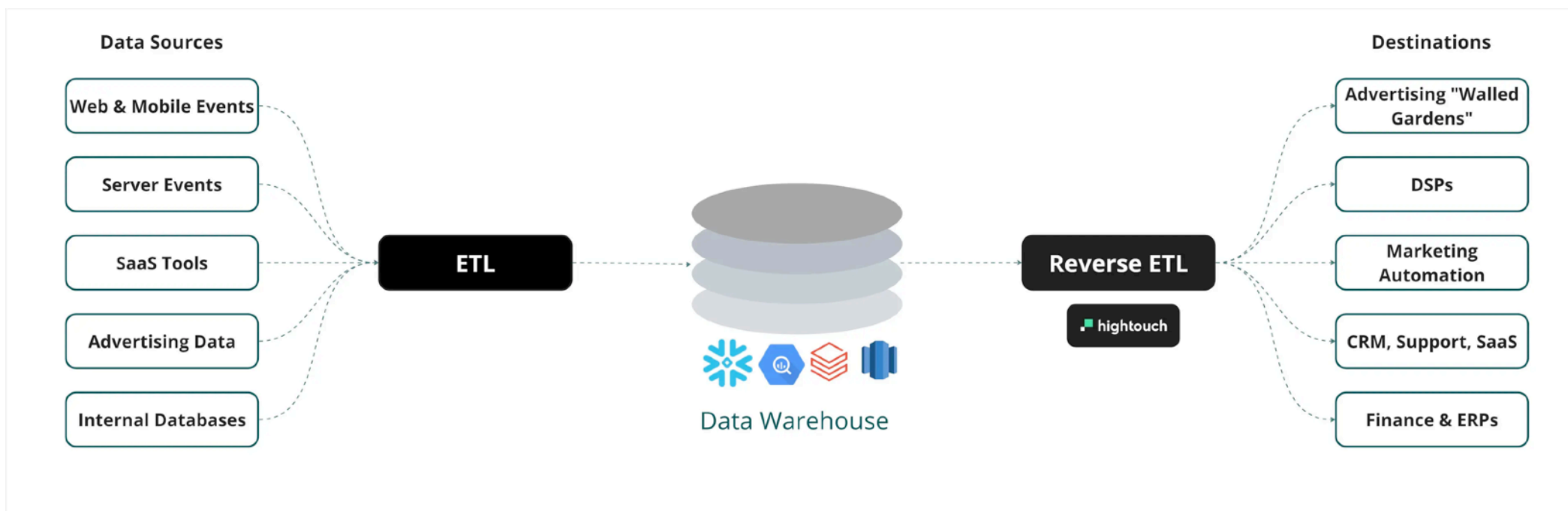
Architectural Design: ETL



DATA WAREHOUSE

Architectural Design: Reverse ETL

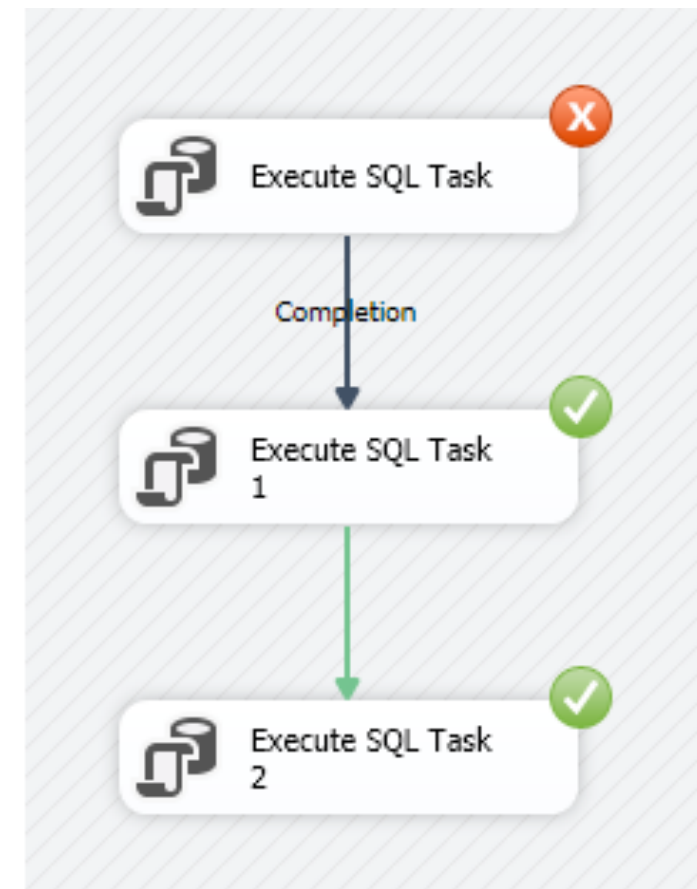
- Reverse ETL takes processed data from the output side of the data engineering lifecycle and feeds it back into source systems
- Several vendors have embraced the concept of reverse ETL and built products around it, such as Hightouch and Census



DATA WAREHOUSE

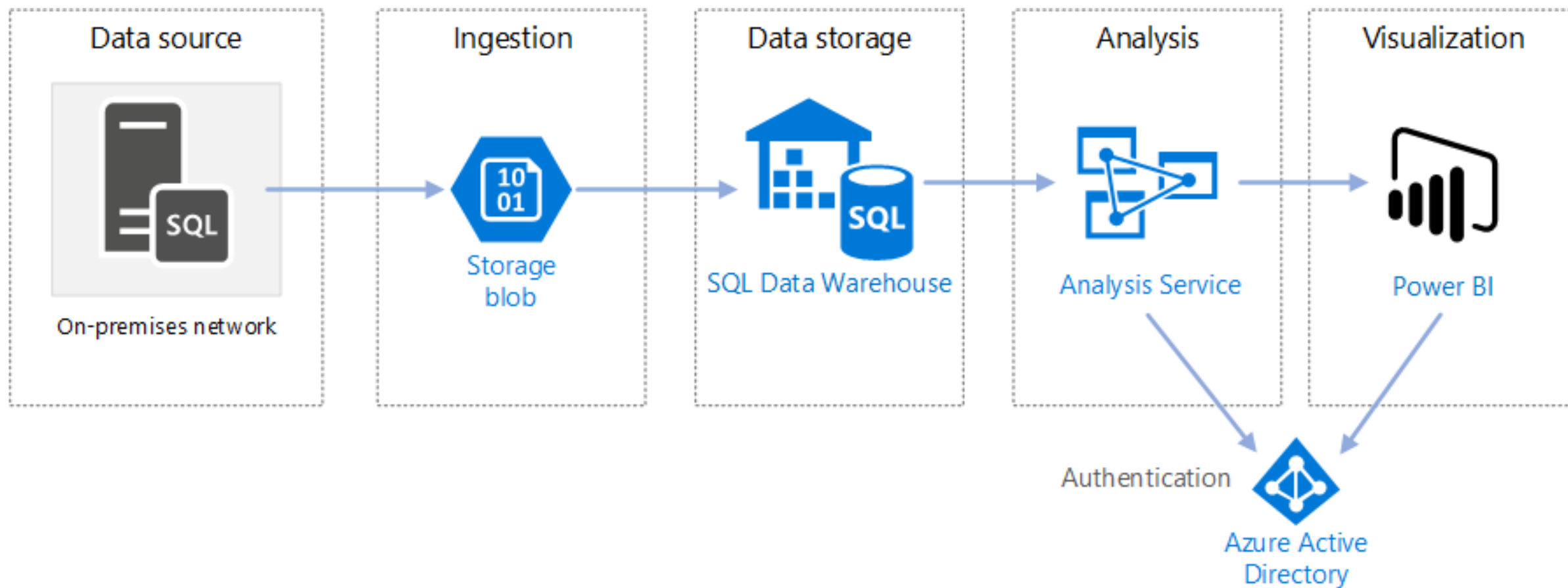
Architectural Design: Management & Control

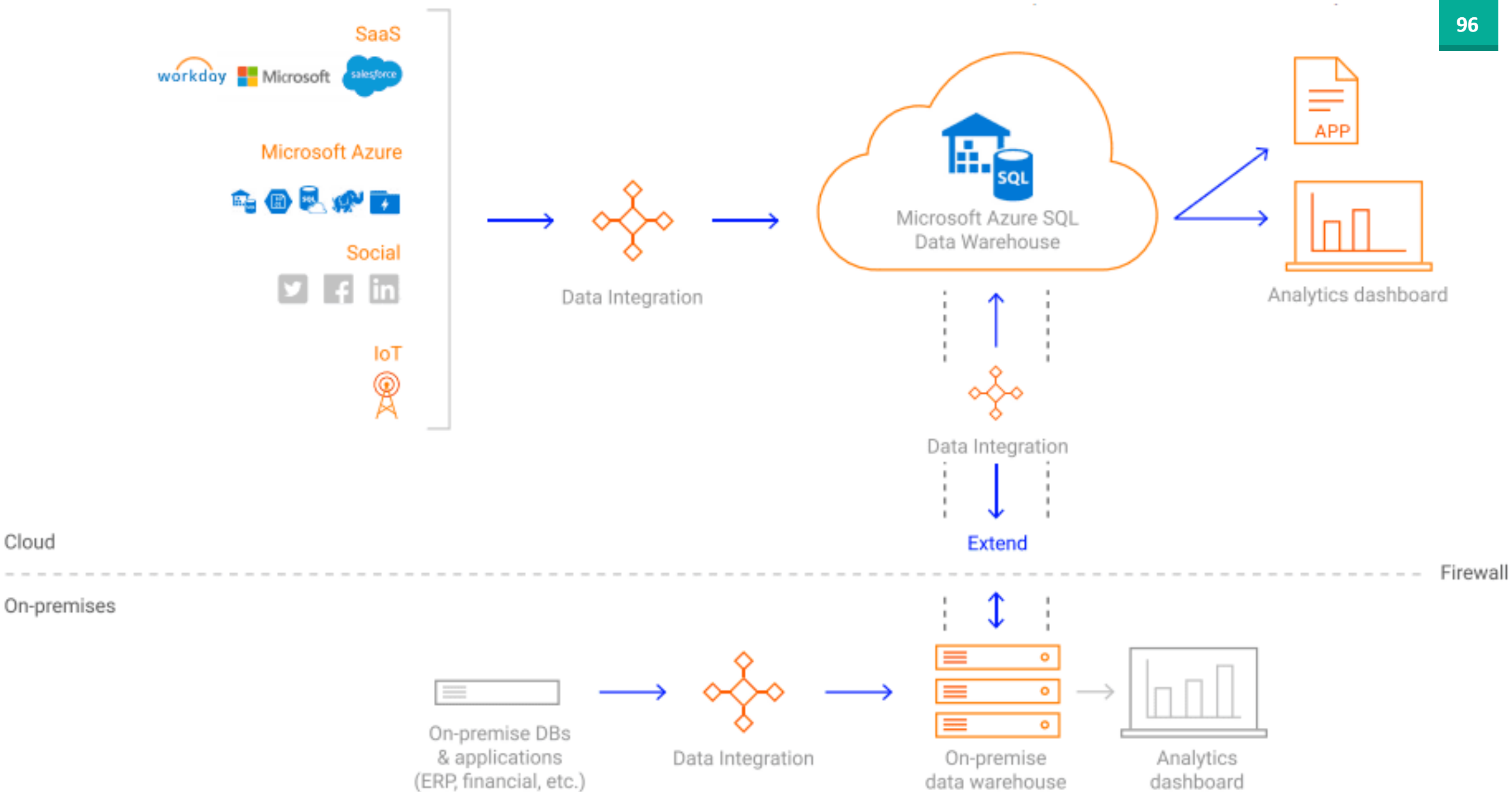
- Management & control module to constantly **monitor** all the ongoing operations, and to step in and **recover from problems** when things go wrong
- Backing up significant parts of the data warehouse and recovering from failures
- Monitoring the growth and periodically archiving data from the data warehouse
- Governing data security and providing authorized access



DATA WAREHOUSE

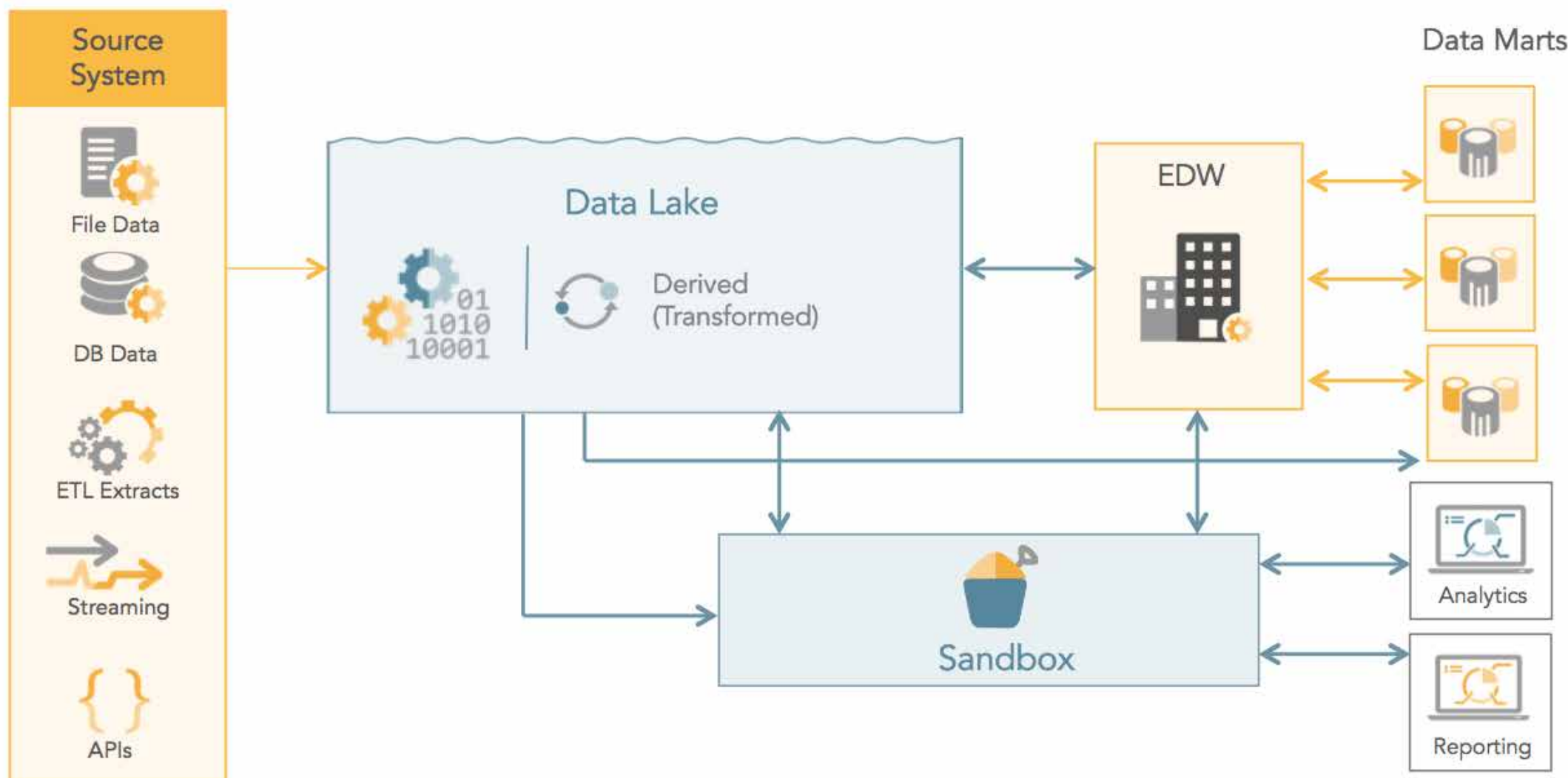
Architectural Design: examples





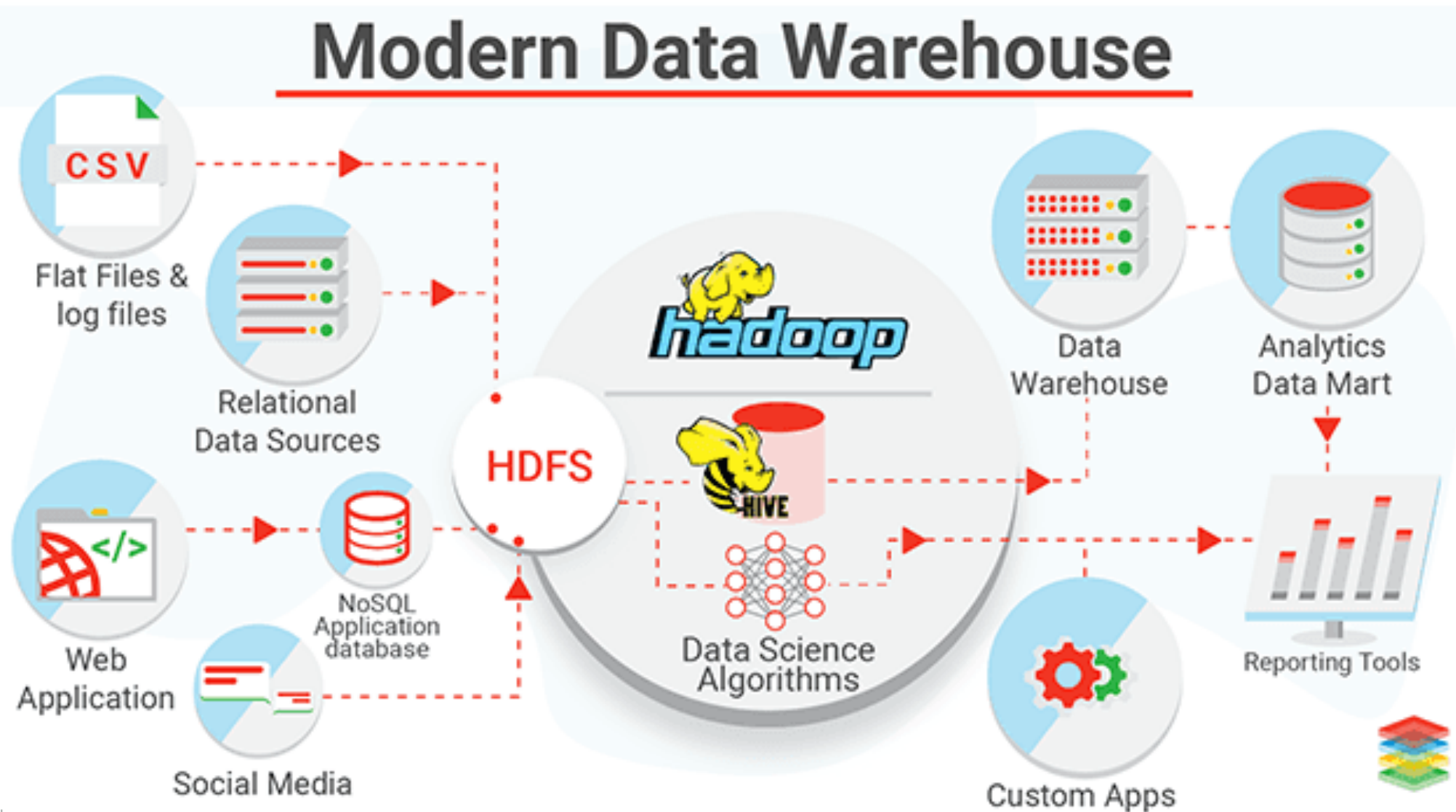
DATA WAREHOUSE

Architectural Design: examples



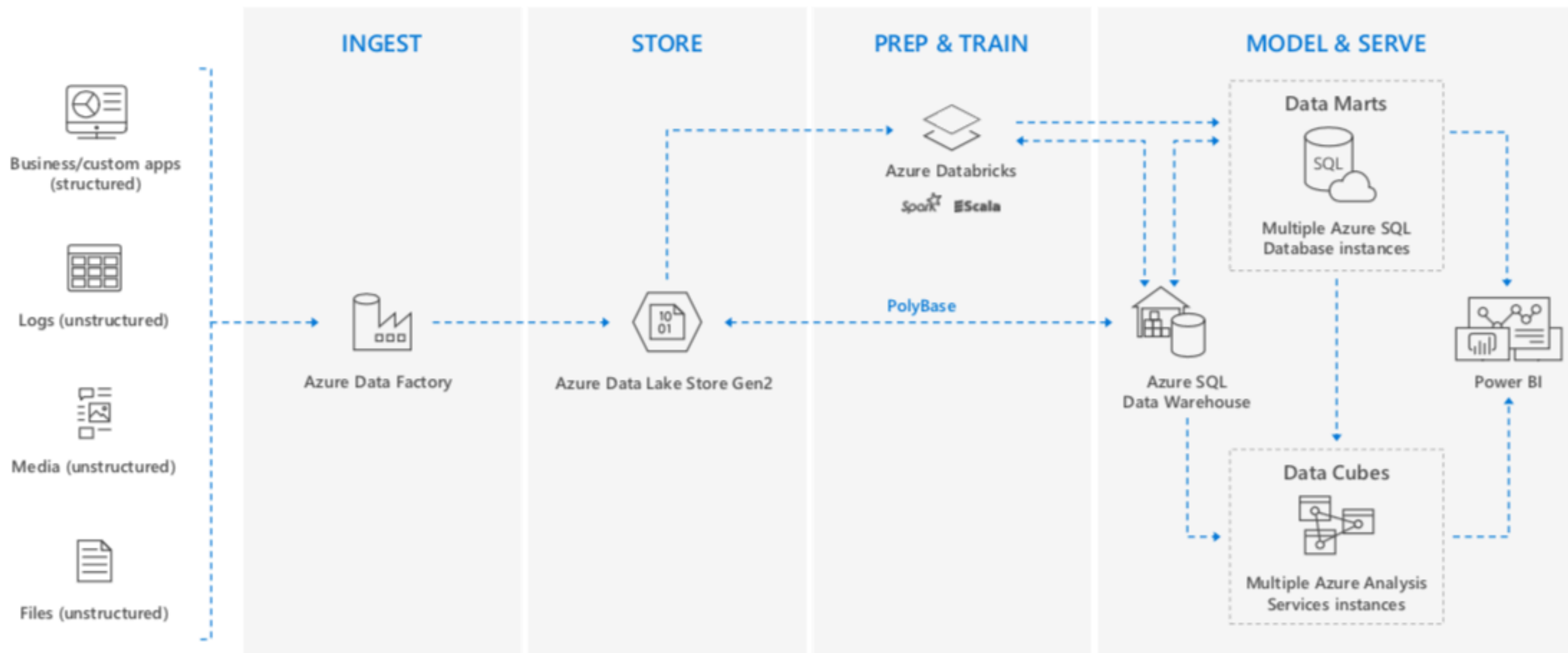
DATA WAREHOUSE

Architectural Design: examples



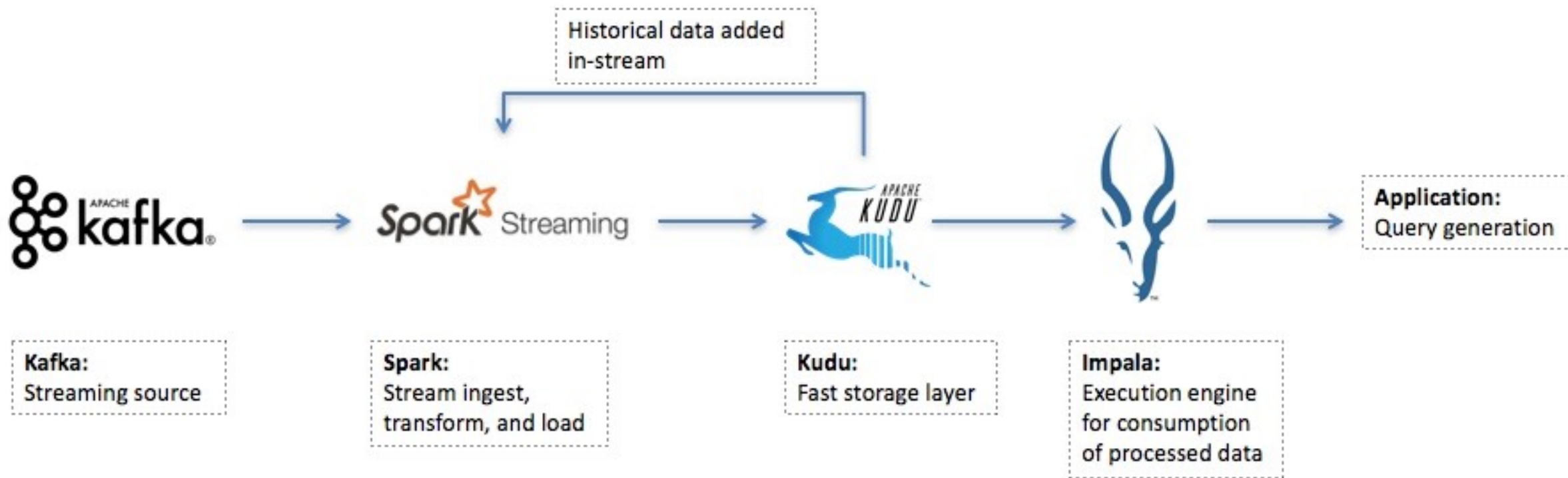
DATA WAREHOUSE

Architectural Design: examples



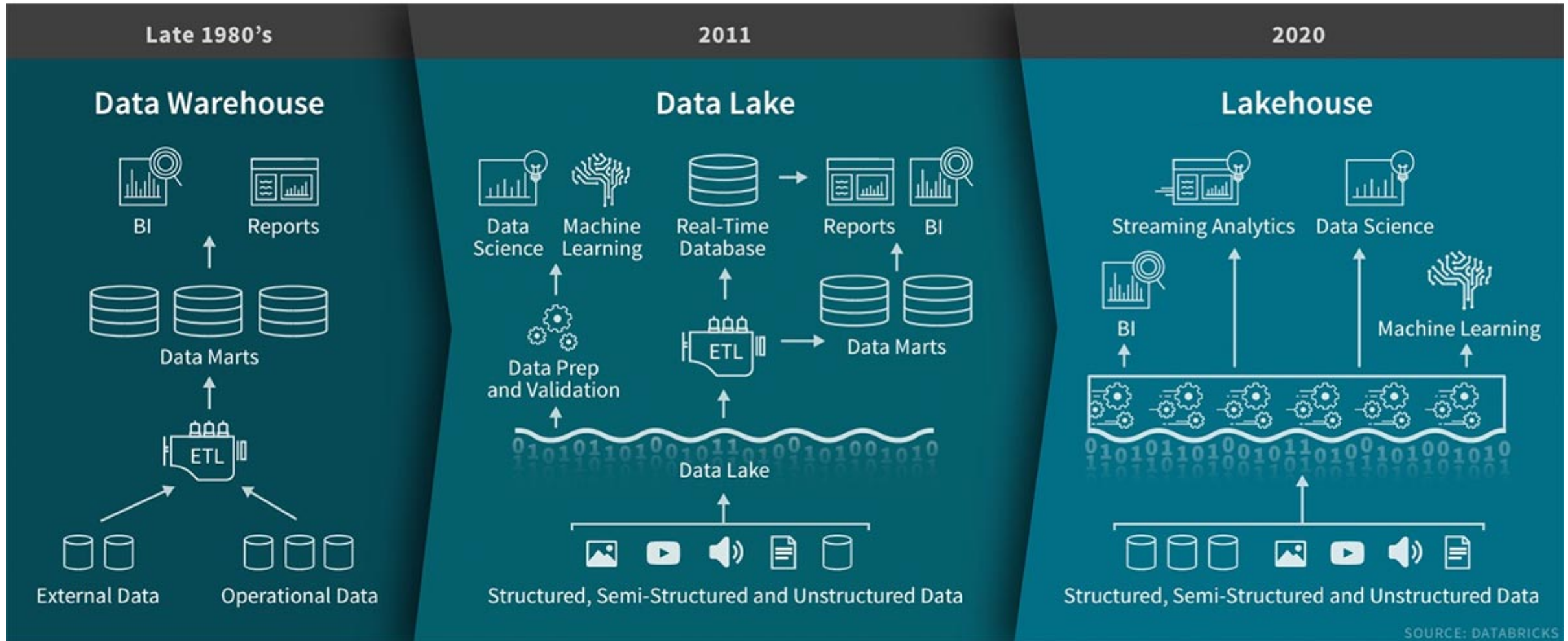
DATA WAREHOUSE

Architectural Design: examples



DATA WAREHOUSE

Architectural Design: examples



AGENDA

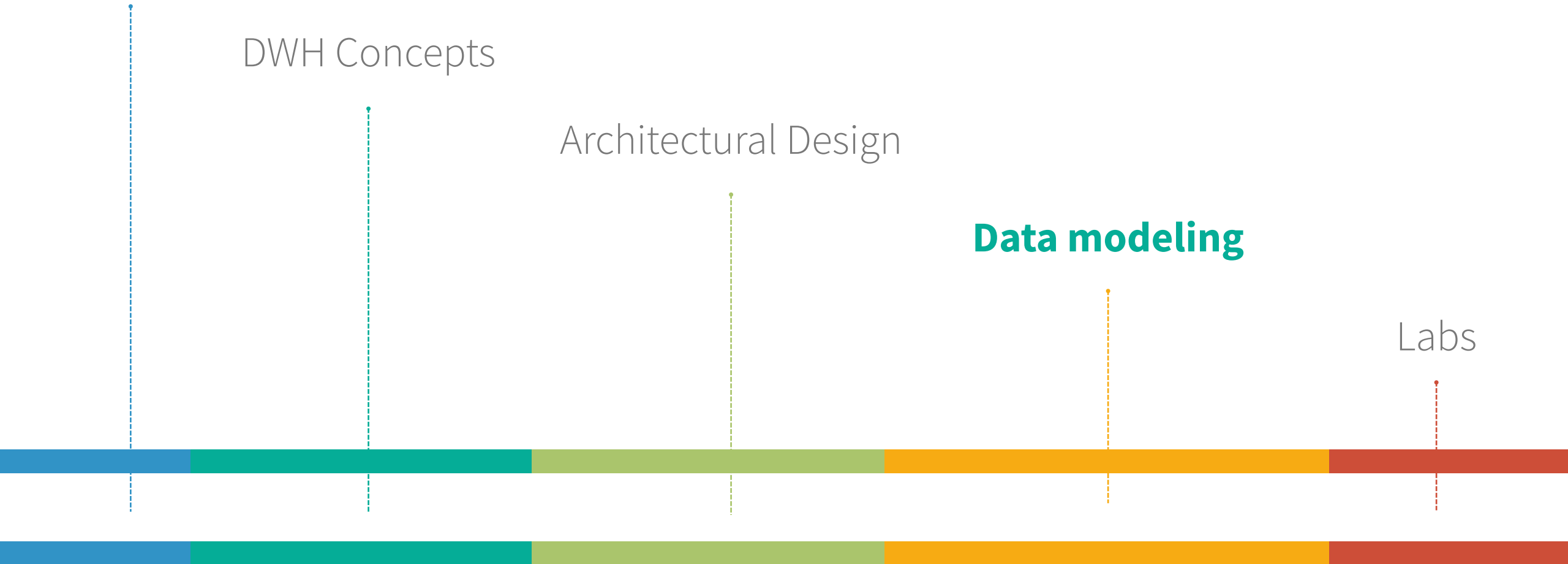
Use Case

DWH Concepts

Architectural Design

Data modeling

Labs



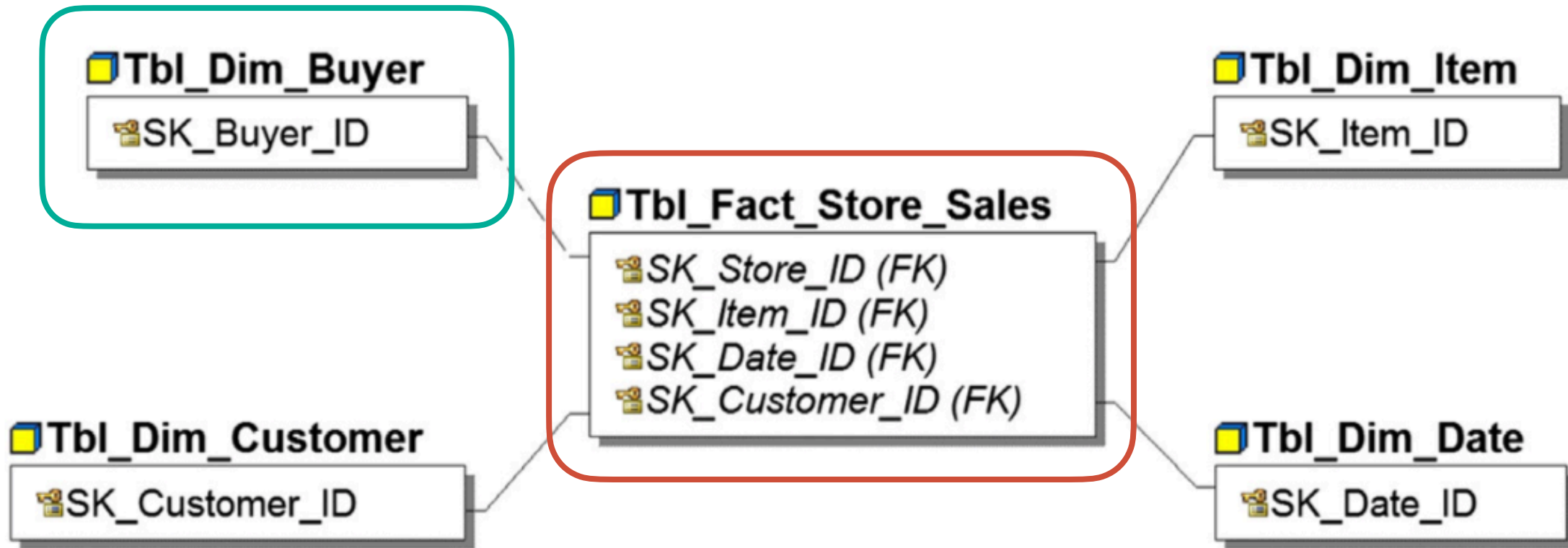
DATA WAREHOUSE

Data modeling

- Like enterprise relationship (ER) modeling for Transactional systems, data warehouse modeling is called **dimensional modeling**
- Dimensional modeling is a logical design technique that specifically fits the needs for BI reporting, query, and analysis
- The key concepts in dimensional modeling are **facts**, **dimensions**, and **attributes**
- Facts, dimensions, and attributes can be organized in several ways, called **schemas**

DATA WAREHOUSE

Data modeling - Dimensional model



DATA WAREHOUSE

Data modeling: Facts

- A fact is a measurement of a business activity (e.g. business event or transaction) and is generally numeric
- Facts provide the measurements of how well or how poorly the business is performing, hence they are referred to as organizational performance measures
- Facts can be aggregated or derived. For example, you can sum up the total revenue or calculate the profitability of a set of sales transactions
- Fact tables are normalized and contain little redundancy. Key concern: minimize, standardize, and make data consistent

DATA WAREHOUSE

Data modeling: Fact table

- A fact table is composed of two types of columns: **keys** and **measures**
- The keys are foreign keys (FK) that point to the primary keys of dimensional tables that are associated with this fact table to enable business analysis
- The relationships between fact tables and the dimensions are one-to-many
- The primary key of a fact table is typically a multipart key consisting of the combination of foreign keys that can uniquely identify the fact table row
- Best practice: foreign keys should never be null

DATA WAREHOUSE

Data modeling: Fact table

- Sometimes, no combination of foreign keys creates the uniqueness required for creating a primary key:
 - Create a surrogate key (auto-incremented)
 - Add a key from the transactional system to the fact table (degenerative dimension)
- The measures in the fact table have a **grain**, which is the level of detail in the measurement of an event
- **Granularity** is determined by its data source

DATA WAREHOUSE

Data modeling: Fact table

Age

18-40	40+
-------	-----

Age, Region

18-40, North	40+, North
18-40, South	40+, South

Age, Region, Income

18-40, North, <\$30k	40+, North, <\$30k
18-40, South, >\$30k	40+, South, >\$30k
18-40, North, >\$30k	40+, North, >\$30k
18-40, South, <\$30k	40+, South, <\$30k

Granularity

-

+



DATA WAREHOUSE

Data modeling: Fact table

- A fact table generally includes many measures.
- All of these measures are related to the business event that the fact represents and they have a level of granularity related to that event
- All rows in the fact table must have a **single, uniform grain**
- It's an accepted best practice to store the lowest, or the most detailed grain captured in the business process

Fact_OnlineSales	
PK	DateKey
PK	StoreKey
PK	ProductKey
PK	CustomerKey
	CurrencyKey
	PromotionKey
	SalesOrderNumber
	SalesOrderLineNumber
	SalesQuantity
	SalesAmount
	ReturnQuantity
	ReturnAmount
	DiscountQuantity
	DiscountAmount
	TotalCost
	UnitCost
	UnitPrice
	DW_ETLLoadID
	DW_LoadDate
	DW_UpdateDate

DATA WAREHOUSE

Data modeling: Type of Facts

- There are three types of measures: additive, semi-additive, and non-additive. The distinction help define how data can be aggregated or summarized
- Additive facts are measures that can be added across all dimensions. e.g. Quantity of items sold, number of customers etc.
- Semi-Additive facts can be added across some dimensions but not others. e.g.bank account balances, the number of students attending a class, or inventory levels
- Non-Additive facts are measures in fact tables that can't be added across any dimensions. e.g. unit prices, ratios, and temperatures

DATA WAREHOUSE

Data modeling: Type of Facts

- A fact table contains fewer attributes than a dimension table (10 attributes or less) but number of records is very large in comparison
- We say that fact tables are **Deep**, not wide
- Fact tables store **sparse data**. This means there will be gaps. Not all combinations of dimension table attributes, values will have their corresponding rows in the fact table

DATA WAREHOUSE

Data modeling: Type of Fact tables

- There are three types of fact tables:
 - **Transaction fact tables** are the most common in dimensional modeling. They have the smallest granularity
 - **Periodic fact tables** record snapshots of data for specific time
 - **Accumulating fact tables** store a record for the entire lifetime of the event, showing the activity as it progresses

DATA WAREHOUSE

Data modeling: Type of Fact tables

Property	Transaction	Periodic	Accumulating
Grain	One row per transaction	One row per time period	One row per lifetime of an event
Date dimension	Lowest level of granularity	End-of-period granularity	Multiple per row
Number of dimensions	Least	Average	Most
Facts	Transaction related	Period related	Numerous events over lifetime
Measurement	Additive	Not additive, average	Need to derive
Conforming dimensions	Yes	Yes	Yes
Conforming facts	Yes	Yes	Yes
Database size	Largest	Smaller	Smallest

DATA WAREHOUSE

Data modeling: Type of Fact tables

- Factless Fact table do not include a numerical measurement, but rather record a set of dimensional entities coming together at a moment in time
- Factless Fact table can also be used to analyze what didn't happen

